0580_w20_qp_42_Q5

5 Ahmed sells different types of cake in his shop.
The cost of each cake depends on its type and its size.

Every small cake costs \$$x$ and every large cake costs \$$(2x + 1)$.

(a) The total cost of 3 small lemon cakes and 2 large lemon cakes is \$12.36 .

Find the cost of a small lemon cake.

\$ ..... [3]

(b) The cost of 18 small chocolate cakes is the same as the cost of 7 large chocolate cakes.

Find the cost of a small chocolate cake.

\$ ..... [3]

(c) The number of small cherry cakes that can be bought for \$4 is the same as the number of large cherry cakes that can be bought for \$13.

Find the cost of a small cherry cake.

\$ ..... [3]

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(d) Petra spends \$20 on small coffee cakes and \$10 on large coffee cakes.
The total number of cakes is 45.

Write an equation in terms of x.
Solve this equation to find the cost of a small coffee cake.
Show all your working.

\$ ..... [7]

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Question	Answer	Marks	Partial Marks
5(a)	1.48	**3**	**B2** for $7x + 2 = 12.36$ or better or **M1** for $3x + 2(2x + 1)$ [= 12.36] or better
5(b)	1.75 or $1\frac{3}{4}$	**3**	**B2** for $18x - 14x = 7$ or better or **M1** for $18x = 7(2x + 1)$
5(c)	[0].8 oe	**3**	**B2** for $4(2x + 1) = 13x$ or **M1** for $\frac{4}{x} = \frac{13}{2x+1}$ oe or correct equation to find number of cakes
5(d)	$\frac{20}{x} + \frac{10}{2x+1} = 45$ oe	**M2**	**B1** for $\frac{20}{x}$ seen or $\frac{10}{2x+1}$ seen
	$90x^2 - 5x - 20$ [= 0] oe	**B2**	**B1** for $\frac{20(2x+1)+10x}{x(2x+1)} = 45$ or better
	$(9x + 4)(2x - 1)$ [= 0] or for $\frac{--1 \pm \sqrt{(-1)^2 - 4(18)(-4)}}{2(18)}$ oe	**M2**	**M1** for factorisation that gives two correct terms when expanded or for correct discriminant or correct $\frac{-b}{2a}$ provided quadratic formula is in correct form
	[0].5 or $\frac{1}{2}$ final answer	**B1**	